

Full Stack Development Project

Documentation

1. Introduction

- **Project Title:** Import Data using Transform Maps (Spreadsheet)
- **Team Members:**
 - **Afrin A:** Project Lead & Target Table Configuration
 - **Archana N:** Staging Set & Field Mapping Lead
 - **Esakkithal E:** Data Transformation & Coalesce
 - **Nardhana T K:** Validation & Re-import Testing
 - **Shifana M A(Lead):** Reporting Engine & Dashboards Lead

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2. Project Overview

- **Purpose:** The primary purpose of this project is to automate and streamline the ingestion of employee datasets from external spreadsheets into ServiceNow. By establishing an Import Set staging table and configuring Transform Maps with Coalesce logic, the project eliminates manual data entry errors and prevents duplicate record creation upon repeated file uploads.
- **Features:**
 - **Staged Data Ingestion:** Staging table (u_employee_import) isolates uploaded raw Excel data before permanent storage.
 - **Automated Field Mapping:** Auto-matching mapping rules between incoming spreadsheet columns and target table attributes (u_employee_test).
 - **Deduplication Engine:** Coalesce rule configured on Employee ID to automatically update existing profiles or insert new ones.
 - **Real-Time Visual Dashboards:** Integrated analytical pie charts and tabular list views consolidated on an Employee Analytics Dashboards page.

3. Architecture

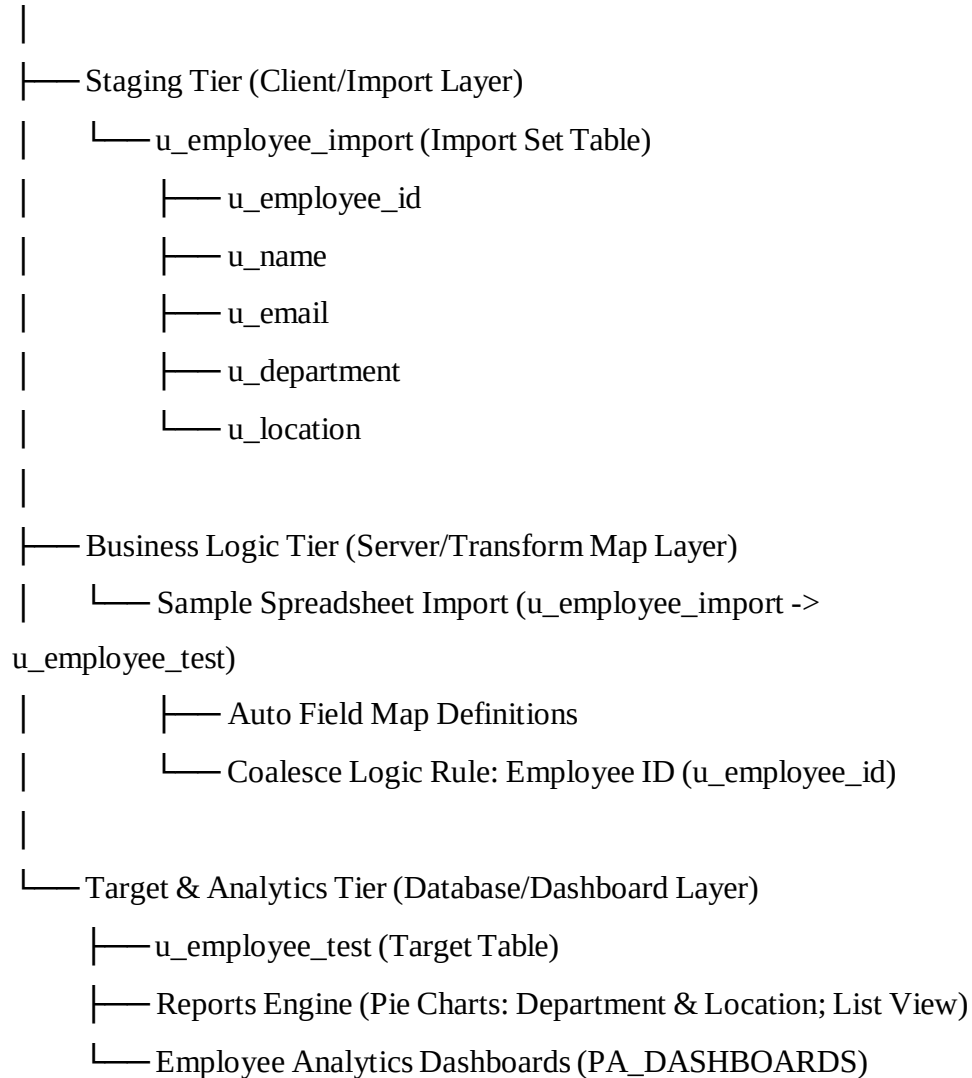
- **Frontend:** Built using the native ServiceNow Service Portal / Workspace interface, rendering dynamic UI forms, report widgets, and Performance Analytics dashboards.
- **Backend:** ServiceNow Application Platform engine handling Import Set processing, Transform Map execution, scriptable business logic, and automated Coalesce checks.
- **Database:** ServiceNow relational database architecture:
 - **Staging Table (u_employee_import):** Temporary table storing uploaded row data.
 - **Target Table (u_employee_test):** Production table containing core attributes (u_employee_id, u_employee_name, u_email, u_department, u_location).

4. Setup Instructions

- **Prerequisites:**
 - Admin access to a ServiceNow Developer Instance.
 - Spreadsheet software (Microsoft Excel / Google Sheets) supporting .xlsx uploads.
- **Installation:**
 1. Log in to the ServiceNow instance with administrator credentials.
 2. Navigate to **System Definition > Tables** and create target table u_employee_test with custom string fields (Employee ID, Name, Email, Department, Location).
 3. Navigate to **System Import Sets > Load Data**.
 4. Upload Sample Spreadsheet.xlsx and target staging table u_employee_import.
 5. Access **System Import Sets > Transform Maps**, map source columns to target fields using **Auto Map Matching Fields**, and set **Coalesce** to True for Employee ID.

5. Folder Structure

ServiceNow Application Scope / Data Schema



6. Running the Application

In a ServiceNow platform environment, execution is handled through native platform operations rather than console commands:

- **Data Load & Transformation Execution:** Navigate to **System Import Sets > Load Data**, select file, then click **Run Transform**.
- **Dashboard Access:** Navigate to **Self-Service > Dashboards** or open form PA_DASHBOARDS.FORM to launch Employee Analytics Dashboards.

7. API Documentation

ServiceNow automatically exposes REST API endpoints for custom tables:

1. Ingest Data Staging

- **Endpoint:** POST /api/now/table/u_employee_import
- **Description:** Inserts raw spreadsheet record into staging table.
- **Request Body Example:**

```
{  
  "u_employee_id": "SB-0004",  
  "u_name": "Ajay",  
  "u_email": "test18@gmail.com",  
  "u_department": "HR", "u_location":  
  "Coimbatore"  
}
```

- **Response:** 201 Created

2. Fetch Employee Records

- **Endpoint:** GET /api/now/table/u_employee_test
- **Description:** Queries processed employee profiles.
- **Response Body Example:**

```
{  
  "result": [  
    {  
      "u_employee_id": "SB-0004",  
      "u_employee_name": "Ajay", "u_email":  
      "test18@gmail.com", "u_department": "IT",  
      "u_location": "Coimbatore"  
    }  
  ]  
}
```

}

8. Authentication

- **Session & Role Management:** Access to Import Set tables and Transform Map modules is governed by ServiceNow's standard **Role-Based Access Control (RBAC)**.
- **Admin Override:** Administrative roles (admin, import_admin) are required to configure transform rules, edit access control lists (ACLs), and register dashboard definitions via PA_DASHBOARDS.FORM.

9. User Interface

- **Form Layout Interface:** Configured form view for u_employee_test displaying all mapped employee details cleanly.
- **Transform Map Field Mapping UI:** Interface displaying side-by-side mapping between source Excel columns and target fields with Coalesce = True highlighted.
- **Analytics Dashboard Page:** Visual page presenting pie chart widgets (Employees by Department, Employees by Location) alongside the active Employee List table.

10. Testing

- **Strategy:** Validation was conducted across 3 test cycles to evaluate insertion, modification, and duplicate prevention:
 1. **Initial Upload:** Validated target table creation and population from base Excel rows.
 2. **Coalesce Update/Insert Test:** Uploaded a file containing 2 modified existing rows and 2 new rows. Verified Transform History logged **2 Inserted** and **2 Updated**.
 3. **Duplicate Ignore Test:** Re-uploaded identical dataset. Verified Transform History logged **0 Inserted**, **0 Updated**, and **4 Ignored**.

11. Screenshots or Demo

- **Staging Set Status:** Verified as "Completed" under Import Sets.
- **Transform Map Config:** Verified Coalesce = True on Employee ID.
- **Transform Log Outcome:** Confirmed 0 duplicate records were generated during re-runs.
- **Demo Access:** Hosted on ServiceNow Developer Instance (myskillwallet.ai Workspace).

12. Known Issues

- **ACL Dashboard Creation Restriction:** Standard Dashboard navigation menu restricted creation permissions for custom roles.
 - *Workaround:* Resolved by invoking the backend form direct URI PA_DASHBOARDS.FORM under system administrator scope.

13. Future Enhancements

- **Scheduled Ingestion:** Configuring Scheduled Import Sets to pull spreadsheet updates automatically from SFTP storage or cloud endpoints.
- **Transform Field Scripts:** Implementing JavaScript transformation scripts to standardize email formats and capitalize location entries during ingestion.
- **Automated Data Validation Rules:** Rejecting source rows with missing or malformed Employee ID entries prior to staging transformation.

